Life Saving Scissors: A Cutting Breakthrough in Gene Editing

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Over 7 million people in the world live with sickle cell anemia, a potentially fatal disease with symptoms including a weakened immune system and pain in bones and joints. However, thanks to the immune system of bacteria, scientists may finally have a cure. What makes this possible is CRISPR Cas-9 and gene editing technology, an exciting yet complex area of research. Gene editing refers to the process by which scientists can alter DNA by adding, removing, or replacing genetic material (MedlinePlus). Currently, the most effective technique is CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats). CRISPR is based on a naturally occurring process in bacteria that was first discovered in 1987 in Japan (MedlinePlus). Today, the technology is already utilized in many fascinating applications and presents several challenging obstacles.

CRISPR is an immune system-like mechanism in bacteria that enables the recognition and memory of viruses. When a virus enters a bacterium, the cell incorporates a small portion of the virus's DNA into its own DNA, utilizing CRISPR arrays, which serve as a memory bank (MedlinePlus). These help the bacteria if they encounter the same or a similar virus. If infected again, CRISPR produces RNA that acts like a guide. It attaches to the virus and helps the bacteria find and fight it (MedlinePlus).

The first step of the CRISPR process involves sgRNA (simple guide RNA), which guides Cas-9 (a protein that acts as molecular scissors) to the DNA sequence that needs to be altered (Gostimskaya, 2022). Then, Cas-9 breaks open the double helix. Now that the DNA has been cut, the cell can be repaired in various ways. One way is through non-homologous end joining (NHEJ), where DNA essentially rejoins the ends together without a perfect base pair match, resulting in frameshift mutations (one or more nucleotides are inserted or deleted, causing the rest of the sequence to be off) or a stop codon. Another method is HDR (homology-directed repair), where the cell takes the time to accurately match the base pairs, resulting in a slower process. This can only occur during S and G2 phases because the DNA is duplicated, but the cell hasn't divided, creating an available copy of the DNA. Lastly, the tools and systems inside the cell repair the double-strand break (DSB). A DSB is when both strands of the double helix are separated, and the cell repairs it rapidly to prevent chromosomal damage. Single-strand breaks are easily repaired because there is

still a template to create the other strand. This is why Cas-9 is novel and important, as it cuts both strands, allowing the cell to precisely fix the mistakes. Scientists are attempting to utilize these techniques to modify DNA in human cells when it is associated with a disease or mutation (Asmamaw & Zawdie, 2021).

Researchers are currently investigating this technology for the treatment of various diseases. Today, treatments are being tested for conditions such as sickle cell anemia, cystic fibrosis, and hemophilia, which are expected to have a significant impact on the field of medicine (MedlinePlus). These specific diseases are relatively straightforward to treat with this technique because they are caused by point mutations. Similarly, hemophilia can be caused by large deletions. This means that in the DNA sequence, only a single nucleotide is mistaken, and therefore, only a tiny portion of the DNA needs to be treated with CRISPR. This is called gene therapy. It uses in vivo procedures, which involve editing cells inside a patient's tissue. It also employs ex vivo procedures, where cells are extracted, edited, and then returned to the patient (Gostimskaya, 2022).

CRISPR is also being explored for cell therapy, which modifies cells to function differently. This could enable researchers to investigate more complex illnesses, such as cancer. One idea is that by using CRISPR, doctors could potentially remove immune system cells from someone with cancer, alter them using CRISPR technology to enhance their ability to fight and recognize cancer cells, reinsert them, and potentially eliminate the tumor (Licholai, 2018).

While CRISPR holds promise, it also presents some challenges. Editing is complex, and errors may occur, such as inserting cells incorrectly or editing the wrong part of DNA, which can cause harmful side effects, including cancer. Additionally, there are numerous unknown factors regarding the long-term effects. While the sickle cell treatment was FDA-approved in December 2023, many clinical trials for other types of CRISPR applications are still awaiting more data, which can take time. Furthermore, CRISPR treatment is expensive, costing over $2 million per patient. So far, gene editing has been primarily used in somatic cells (body cells). Ethical questions arise about using this in germ cells—egg and sperm cells—or embryos because the effects can be passed down through generations (MedlinePlus). The US Government currently does not allow federal funds for gene editing research in germ cells. Thus, when CRISPR technology becomes more advanced, should it be used only for medical situations or to enhance appearance and traits? Should we be using CRISPR on plants and animals, and if so, how would that affect ecosystems?

Overall, CRISPR is a cutting-edge technology with many applications. While it is primarily used in clinical trials, it holds considerable promise for the future as a potential treatment and even cure for molecular genetic diseases with a simple underlying cause.